CHAPTER 20

Spinning-satellite control-system design

20.1. Introduction

This chapter describes how to create a preliminary control-system design for a spinning spacecraft. In this case, we will look at a satellite that is spin-stabilized in transfer orbit.

20.2. Spinning-spacecraft operation

Acquisition is the transition between transfer orbit and mission orbit. In transfer orbit, the spacecraft is spinning about its solid motor axis at the attitude needed for the delta-V burn to put the satellite in the geosynchronous orbit, or a nearby orbit from which the satellite can drift to the station. With this design, we have several options. If we add a Sun sensor we can

1. Acquire the Sun;
2. Rotate about the sunline (a safe attitude because the spacecraft is gyroscopically stable and the arrays can be easily pointed at the Sun to deliver power);
3. Acquire the Earth;
4. Lock on to the Earth;
5. Spin up the momentum-wheel assembly (MWA).

This is known as a Sun/Earth acquisition (SEA). This is desirable because the Earth is a small target at geosynchronous altitudes. Even without a Sun sensor, we can use the solar arrays as a Sun sensor of sorts. Another option is to

1. Precess the spin axis into the orbit plane;
2. Rotate in yaw until the desired orientation is achieved;
3. Spin up the MWA;
4. Lock onto the Earth.

A third option is to

1. Precess the spin-axis to orbit normal (leaving the MWA axis in the orbit plane);
2. Spin up the MWA. The spacecraft will rotate 90 degrees since the inertial momentum vector cannot be moved by internal torques;
3. Lock onto the Earth.

This last option is known as the dual-spin turn and was pioneered by RCA Astro Electronics. This last choice does not require any inertial reference after the precession to orbit normal is complete. Measuring the spin-axis attitude before the dual-spin-turn is done using horizon sensors and a single-axis Sun sensor.

ADCS - Spacecraft Attitude Determination and Control
https://doi.org/10.1016/B978-0-32-399915-1.00032-2

When choosing an acquisition scheme one must pay close consideration to power. During parts of the acquisition, the solar arrays may not get much Sun and if the acquisition takes too long it could leave the spacecraft in a perilous state. Since the spacecraft is a major-axis spinner, the last option does not require any additional sensing.

20.3. Transfer orbit

There are several choices for delta-V engines:

1. Solid motor (big thrust 1000s of N);
2. Liquid-fuel motor (moderate thrust 100s of N);
3. Electric thrusters (low thrust $\ll 1$ N).

High-thrust engines usually require that the spacecraft be spun for stability during the burn. Spacecraft with moderate thrust levels can be spun or three-axis stabilized. The cheapest option for small satellites is to use a solid motor. This design will use a solid motor.

When it comes time to fire the solid motor, the satellite will need to be pointing in the right direction. For that, the operators need attitude information and thrusters to adjust the orientation. At a minimum, it is necessary to know the inertial orientation. Usually, the Sun or Earth is used as a reference. Since the spacecraft is spinning, the spacecraft spin becomes the scanning mechanism. The cheapest sensors are single-axis Sun sensors and horizon sensors.

Thrusters are needed for station keeping, which means a set of thrusters that can provide three-axis control. It is undesirable to add additional thrusters for transfer orbit, therefore it is necessary to make certain that the thrusters are usable (i.e., are not blocked by undeployed solar arrays and antennas) for controlling the spacecraft spin rate or changing the spin-axis orientation.

Since we have chosen a spinner, the final consideration is whether the spacecraft is spinning about its major or minor axis. If the former is true we are done with the preliminary design. Otherwise, we will need a control system to damp nutation and prevent the spacecraft from spinning about its major axis.

20.4. Spinning transfer orbit

20.4.1 Dynamics

A typical spinning spacecraft in transfer orbit is composed of a fairly rigid (especially compared to the deployed configuration) core spacecraft with a significant fraction of its mass in the form of liquid propellant. Fuel motion will cause energy dissipation in the spacecraft. If the spacecraft is a major-axis spinner this generally does not pose a problem since any energy dissipation will cause the spacecraft to return to its major-axis spin state. However, if the spacecraft is a minor-axis spinner its attitude will diverge and it will attempt to spin about its major axis. Consequently, minor-axis spinners require nutation-control systems. In this example, the spacecraft will be assumed to be a major-axis spinner, and nutation control will not be discussed further.

The equations of motion for a rigid spinning spacecraft are

$$I\dot{\omega} + \omega^{\times} I\omega = T \tag{20.1}$$

where I is the spacecraft inertia, which is assumed to be constant, and ω is the spacecraft angular velocity with respect to the inertial frame measured in the body frame. If the spacecraft is spinning about the z-axis with a constant rate ω_o the transfer functions of the linearized plant are

$$\begin{bmatrix} \omega_x \\ \omega_y \end{bmatrix} = \frac{\begin{bmatrix} s & k_x \\ -k_y & s \end{bmatrix}}{s^2 + k_x k_y} \begin{bmatrix} T_x/I_x \\ T_z/I_z \end{bmatrix} \tag{20.2}$$

$$k_x = \frac{\omega_o(I_z - I_y)}{I_x} \tag{20.3}$$

$$k_y = \frac{\omega_o(I_z - I_x)}{I_y} \tag{20.4}$$

These reduce to a pair of single integrators if ω_o, the z-axis spin rate, is zero.

The inertial angular-momentum vector is

$$H = AI\omega \tag{20.5}$$

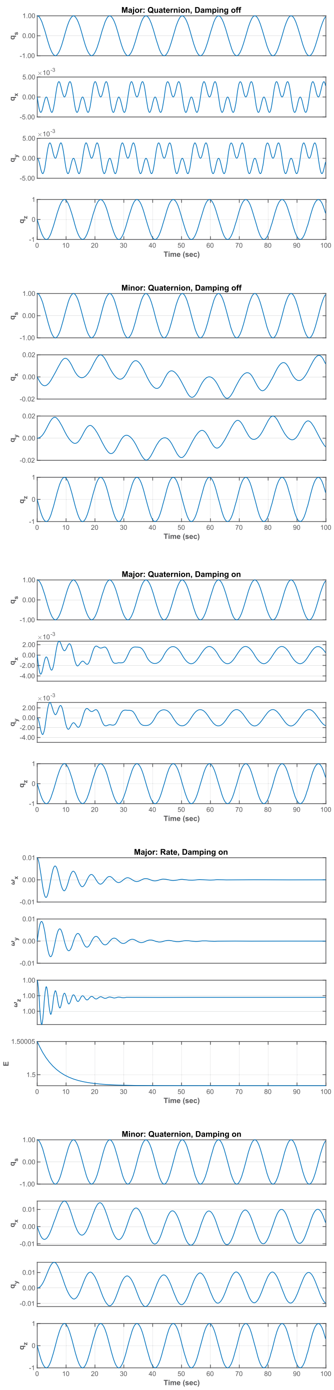
where A transforms a vector from the body frame to the inertial frame and the energy is

$$E = \frac{1}{2}\omega^T I\omega \tag{20.6}$$

H is fixed in magnitude and direction in the absence of external torques. Internal torques can cause E to change even though H does not. For example, when a minor-axis spinner has energy dissipation it will reorient itself so that its energy is a minimum. Even though the spin-axis changes, the angular momentum does not and it redistributes itself in the body frame so that it remains fixed in direction and magnitude in the inertial frame.

The rotation of the spin-axis about the momentum vector is often known as precession. Here, precession will mean the motion of the momentum vector with respect to the inertial frame. Since momentum is conserved, only an external torque, such as the torque produced by a thruster, can precess the momentum vector.

Example 20.1 shows the dynamics for a minor- and a major-axis spinner with and without rate damping.




```matlab
%% Nutation damping

minor   = true;
control = true;

if( minor )
  type  = 'Minor';
  inr   = diag([2 3 1]);
else
  type  = 'Major';
  inr   = diag([1 2 3]);
end

dT  = 0.1;
n   = 1000;
xP  = zeros(8,n);
if( control )
  c = [0.1;0.1;0];
  damping = 'on';
else
  c = [0;0;0];
  damping = 'off';
end

x = [1;0;0;0;0.01;0;1];
for k = 1:n
  omega    = x(5:7);
  e        = 0.5*omega'*(inr*omega);
  xP(:,k)  = [x;e];
  x        = RK4(@RHS,x,dT,0,inr,c);
end

t = (0:n-1)*dT;
yL = {'q_s' 'q_x' 'q_y' 'q_z'...
  '\omega_x' '\omega_y' '\omega_z' 'E' };

s = sprintf('%s: Quaternion, Damping %s',type,damping);
TimeHistory(t,xP(1:4,:),yL(1:4),s);
s = sprintf('%s: Rate, Damping %s',type,damping);
TimeHistory(t,xP(5:8,:),yL(5:8),s);

%% Dynamics right-hand-side
function xDot = RHS(x,~,inr,c)
q          = x(1:4);
omega      = x(5:7);
omegaDot = -inr\(c.*omega + cross(omega,inr*omega));
xDot       = [QIToBDot(q,omega);omegaDot];
end
```

Example 20.1: Nutation of a minor- and major-axis spinner with and without damping.

20.4.2 Actuators and sensors

Thrusters are used to control the orientation and spin rate of the spacecraft during transfer orbit. Fixed-pulsewidth, throttleable, or pulsewidth-modulated thrusters can be used. Fixed pulsewidth is the least flexible. Both pulsewidth modulation and throttling permit the use of linear control laws, but both have threshold and saturation nonlinearities associated with them. Keep in mind that the propellant-pressurization system will affect the use of thrusters. This spacecraft employs pulsewidth modulation.

Sun sensors and horizon sensors are used for attitude determination. The Sun sensor has two outputs. Each time the Sun passes through the Sun-sensor boresight/spin-axis plane, one provides a pulse. This provides a convenient timing reference for computing spin rate. The other outputs the Sun angle in the spin-axis/boresight plane. Gyros are available, but the spin rates during transfer orbit may saturate the gyros.

The horizon sensors detect the presence of the Earth. The horizon sensors view the Earth in the CO_2 band, around 14 μm. The Earth's atmosphere is relatively stable at this frequency. When the Earth is in the sensor field-of-view the output of the sensor jumps. The horizon sensors have circuitry that detects the transition from cold space to warm Earth. There are many ways to do this. The simplest outputs a pulse whenever the output of the detector reaches a fixed threshold. Another outputs a pulse when the output reaches a fixed percentage of the peak. The latter tends to be less sensitive to Earth atmospheric-radiance variations.

Horizon sensor measurements have many error sources. These include:

1. Earth seasonal and daily radiance variations;
2. Sun and Moon interference;
3. Sensor dynamics;
4. Misalignments;
5. Irregularities in the Earth's figure;
6. and so on.

The geometry of spin-axis attitude determination is illustrated in Fig. 20.1.

20.4.3 Changing the spin rate

Two ways are available to change the spin rate. One is by manually firing the appropriate thruster(s). The second is to use the automatic spin-rate control function. This function measures spin rate using the Sun-sensor pulses and a spin-rate estimator structure and controls the spin rate with a simple proportional controller. The estimator accounts for spin-rate changes due to firing thrusters.

20.4.4 Spin-axis reorientation

Reorientations of a spinning spacecraft are performed using what is known as a spin-precession maneuver. The idea is to reorient the momentum vector without changing

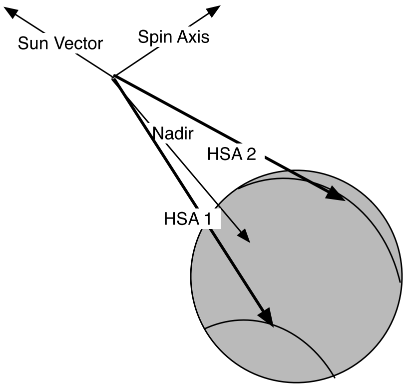


Figure 20.1 Spin-axis attitude determination diagram.

the spin rate. As only the orientation of the spin-axis is of interest, we need only concern ourselves with two angles, right ascension, and declination.

If we have continuous information about the spin-axis attitude, it is possible to perform a great-circle precession. This is illustrated in Fig. 20.2, and the spin-axis moves in the plane represented by the shaded region. α is right ascension and δ is declination. This could be performed if the satellite had three-axis gyros or if it updated its attitude from horizon and Sun-sensor measurements using a Kalman filter. A simpler approach is known as a rhumb-line precession. The geometry is illustrated in Fig. 20.3.

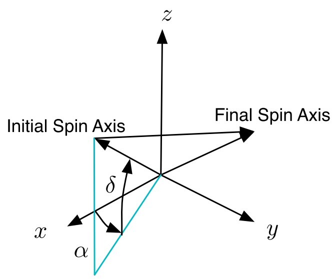


Figure 20.2 Great-circle precession diagram.

The equation for the rhumb-line angle is [1]

$$\tan\psi = \frac{\xi_s(t_f) - \xi_s(t_i)}{\log_e\left[\frac{\tan(\theta_s(t_f)/2)}{\tan(\theta_s(t_i)/2)}\right]} \tag{20.7}$$

where θ_s is the spin-axis declination and ξ_s is the spin-axis azimuth in the Sun plane, t_i is the start time, and t_f is the final time. The angle traversed is

$$\sigma = \begin{cases} |[\theta_s(t_f) - \theta_s(t_i)] \sec \psi| & \psi \neq 90° \\ |[\xi_s(t_f) - \xi_s(t_i)] \sin \theta_s| & \psi = 90° \end{cases} \tag{20.8}$$

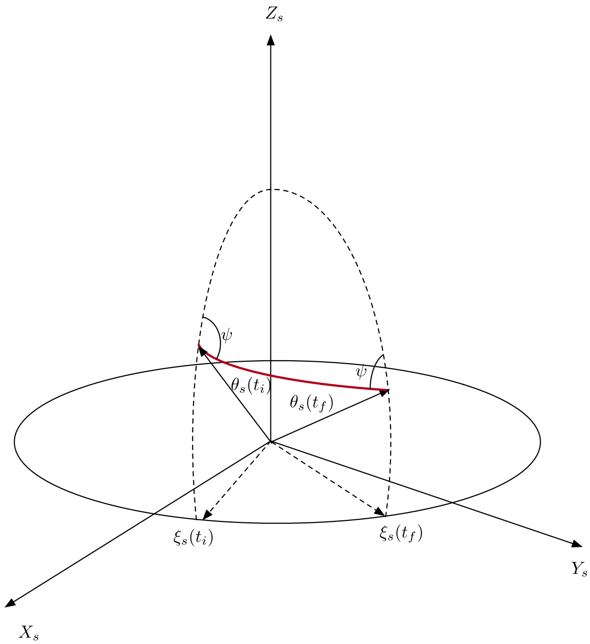


Figure 20.3 Rhumb-line precession diagram.

The rhumb-line algorithm is based on firing thrusters at a fixed time delay from the Sun pulse [1]. The thrusters are fired twice per spin period to control nutation.

The coordinate frame is such that the Sun vector is the $+Z_s$-axis. The initial and final spin-axis vectors are shown as is the path connecting them. The azimuth angle, ψ, which is the angle between the trajectory and the Sun/spin-axis plane, is a constant during the maneuver. ψ is constant because a fixed delay between the Sun crossing the spin-axis Sun-sensor plane and the thruster firing is used.

The first step is to find the initial and final unit vectors for the spacecraft spin-axis, u_i, and u_f. We then create a Sun-oriented coordinate system, where Z_s is the unit Sun vector. The other axes are found by

$$Y_s = Z_s \times u_i \tag{20.9}$$

$$X_s = Y_s \times Z_s \tag{20.10}$$

The vectors are unitized after each step. The transformation matrix from the inertial to the Sun frame is

$$C = \begin{bmatrix} X_s^T \\ Y_s^T \\ Z_s^T \end{bmatrix} \tag{20.11}$$

The transformed vectors are

$$u_{i_s} = Cu_i \tag{20.12}$$

$$u_{f_s} = Cu_f \tag{20.13}$$

The angles are

$$\theta = \cos^{-1} u_{s_z} \tag{20.14}$$

$$\xi = \tan^{-1}\left(\frac{u_{s_y}}{u_{s_x}}\right) \tag{20.15}$$

The arc tangents should use the atan2 function. The rhumb line is computed from Eq. (20.8).

Simulating a spin-precession maneuver requires a very small time step so that the timing of the torque pulses can be accurately computed. For simplicity, assume the spacecraft is spinning about its z-axis and that the sensor boresight is along the $+x$-axis. Then, the Sun pulse happens when the y component of the Sun vector in the body frame changes sign.

We want to fire a thruster to produce an angular rate about an axis perpendicular to the spin-axis and at the angle ψ. This is shown in Fig. 20.4.

The precession rate is based on the applied torque

$$\omega_p = \frac{T}{h} \tag{20.16}$$

where h is the angular momentum and T is the torque. The precession per pulse is based on the arc efficiency. If ω is the spin rate

$$\eta = \frac{\sin\delta}{\delta} \tag{20.17}$$

$$\delta = \frac{\omega\tau}{2} \tag{20.18}$$

This accounts for the torque not being applied exactly at the delay time.

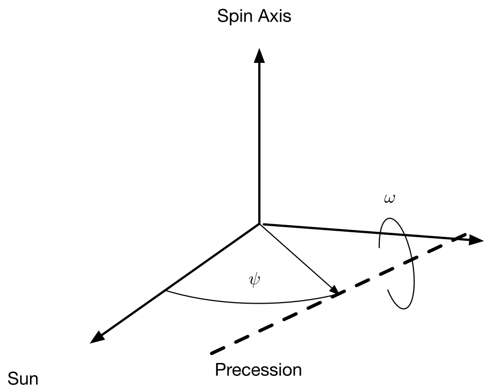


Figure 20.4 Thruster geometry for spin-precession maneuvers.

The angle change per pulse is

$$\delta\theta = \frac{\omega_p \tau}{\eta} \tag{20.19}$$

where τ is the effective pulsewidth, accounting for rise and fall times.

An example maneuver is shown in Example 20.2. The thrusters produce torques about the x-axis of the spacecraft. A pulse is fired with every Sun pulse and then one-half period later. The torque is a one-time-step approximation for the actual thruster pulse that might last for multiple time steps. In a more sophisticated simulation, the pulses would last for several time steps and have a rise and fall time.

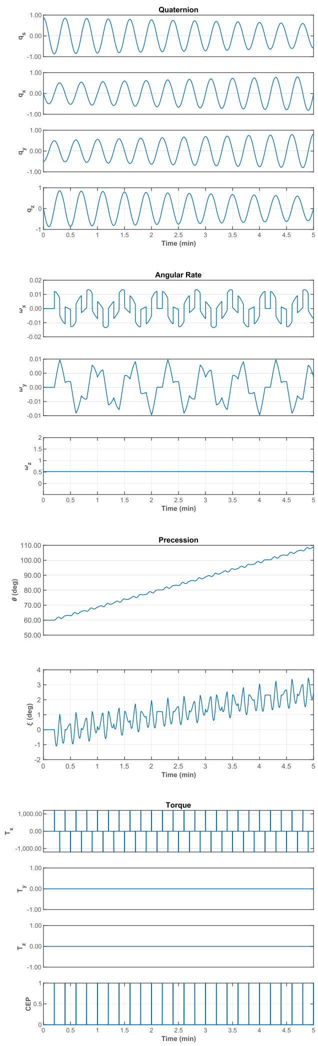
20.4.5 Attitude determination

Attitude determination is performed using three measurements, the Sun angle and the times when the horizon sensor scan crosses the edge of the Earth measured relative to the time at which the Sun crosses the Sun-sensor boresight/spin-axis plane. The three measurements are resolved into the Sun angle, the dihedral angle, and the chordwidth. The chordwidth is the difference between the trailing and leading edge times divided by the spin rate; the dihedral angle is the sum of the leading- and trailing-edge times divided by twice the spin rate. If the noise statistics for the leading- and trailing-edge times are identical, then the dihedral angle and chordwidth are uncorrelated.

Besides the measurements, attitude determination also requires the orbital position of the satellite. Consequently, accurate attitude determination requires accurate orbit determination. The attitude determination geometry is illustrated in Fig. 20.5.

The three angles give an unambiguous attitude reference. Usually, many measurements spaced over a period around apogee are taken and a least-squares algorithm is used to estimate both attitude and any biases in the data.

```
rToD     = 180/pi;
n        = 30000;
dT       = 0.01;
inr      = diag([1e3;1e3;1.3e3]);
omega    = 5*pi/30;
q        = Eul2Q([0;pi/3;0]);
x        = [q;0;0;omega];
period   = 2*pi/omega;
t        = (0:n-1)*dT;
pW       = 2*dT;
tPulse   = period/2;
zS       = Unit([0;0;1]);
uSpin    = [0;0;1];
uI       = QTForm(q,uSpin);
yS       = Unit(cross(zS,uI));
xS       = Unit(cross(yS,zS));
c        = [xS';yS';zS'];
uI       = [1;0;0];
uF       = Unit([1;0;1]);
uSunECI  = zS;
[thetaI,xiI] = SpinAtt(c,uI);
[thetaF,xiF] = SpinAtt(c,uF);
num      = xiF - xiI;
den      = log(tan(0.5*thetaF)) - log(tan(0.5*
           thetaI));
azimuth  = atan2(num,den);
prec     = (thetaI-thetaF)*sec(azimuth);
if ( prec < 0 )
  prec     = - prec;
  azimuth  =   azimuth + pi;
end
yOld     = 0;
tPos     = inf;
tNeg     = inf;
tDelay   = azimuth/period/(2*pi);
xP       = zeros(13,n);
for k = 1:n
  [cEP,yOld] = SSA(x(1:4),uSunECI,yOld);
  torque = [0;0;0];

  % Fire every half period
  if( cEP )
    tPos     = t(k) + tDelay;
    pulsePos = true;
  end

  if( t(k) > tPos && pulsePos )
    tNeg      = t(k) + period/2;
    torque    = [1200;0;0];
    pulseNeg  = true;
    pulsePos  = false;
  end

  if( t(k) > tNeg && pulseNeg )
    torque    = -[1200;0;0];
    pulseNeg  = false;
  end

  [theta,xi] = SpinAtt(c,QTForm(x(1:4),uSpin));
  xP(:,k)    = [x;theta;xi;torque;cEP];
  x          = RK4(@RHS,x,dT,0,inr,torque);
end

yL = {'q_s' 'q_x' 'q_y' 'q_z' '\omega_x' ...
'\omega_y' '\omega_z' '\theta␣(deg)' ...
'\xi␣(deg)' 'T_x' 'T_y' 'T_z' 'CEP'};

k = 1:4;
TimeHistory(t,xP(k,:),yL(k),'Quaternion');
k = 5:7;
TimeHistory(t,xP(k,:),yL(k),'Angular␣Rate');
k = 8:9;
TimeHistory(t,xP(k,:)*rToD,yL(k),'Precession');
k = 10:13;
TimeHistory(t,xP(k,:),yL(k),'Torque');
Figui
%% Dynamics
function xDot = RHS(x,~,inr,torque)
q        = x(1:4);
omega    = x(5:7);
omegaDot = inr\(torque - cross(omega,inr*omega));
xDot     = [QIToBDot(q,omega);omegaDot];
end
%% Angle calculation
function [theta,xi] = SpinAtt(c,uSpin)
uS    = c*uSpin;
theta = acos(uS(3));
xi    = atan2(uS(2),uS(1));
end
%% Sun sensor
function [cEP,y] = SSA(qECIToBody,uSunECI,yOld)
uSunBody = QForm(qECIToBody,uSunECI);
y = uSunBody(2);
if( y > 0 && yOld < 0 )
  cEP = true;
else
  cEP = false;
end
end
```



Example 20.2: Spin-precession maneuver.

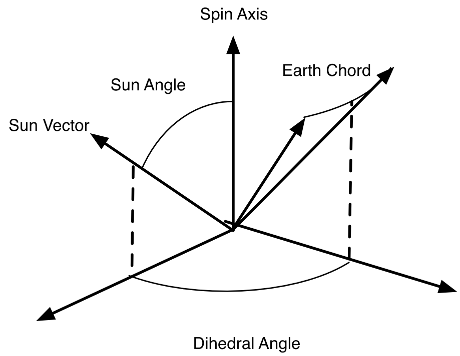


Figure 20.5 Attitude determination geometry diagram.

20.4.6 Delta-V engine firing

The delta-V engine is fired near apogee to put the spacecraft into the geosynchronous orbit. Since the spacecraft is rarely at its station at the time of the firing, the maneuver is planned to leave the spacecraft in a slightly eccentric orbit so that it will drift to its final station. The firing of the AKM causes nutation and a slight spinup of the spacecraft due to center-of-mass misalignments.

References

[1] M.H. Kaplan, Modern Spacecraft Dynamics and Control, Wiley, 1978.